

PRAJWAL TIDKE

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SUMMARY

Senior Data Engineer with **5+ years** architecting production ETL systems processing **1 TB+** daily across AWS, Databricks, and Snowflake for clients managing **\$550B+** in assets. Drove **50%** infrastructure cost reductions and **80%** faster pipeline execution across 6+ enterprise engagements. Currently pursuing MS in Data Science (**4.0 GPA**) at SDSU while building LLM-powered applications with RAG, fine-tuned models, and real-time inference.

EXPERIENCE

San Diego State University | *Research Assistant*

Feb 2025 - Feb 2026

- Designed and built KALM, a mental-health chatbot for construction workers using FastAPI, GPT-4.1, and ChromaDB, adding real-time crisis detection and AES-128 encrypted storage for more than **1,000 clinical knowledge chunks**, which enabled workers to receive immediate assistance and increased response reliability
- Developed prompt engineering strategies for a **5-stage** ALGEE therapeutic framework, handling **53 crisis signal patterns** and four AI personas to improve mental-health support access.
- Implemented KBLaM architecture-constructed **1000+ domain-specific knowledge-base triples**, built QA evaluation datasets, and fine-tuned LLaMA, achieving **12% improvement** in domain-specific accuracy.
- Engineered WGAN-based synthetic data pipeline for biosignal analysis, generating **50K+ samples** and reducing core-temperature prediction **MAE by 15%** over baseline.

TresVista Analytics | *Associate Data Engineer*

Oct 2023 - Aug 2024

- Led a 2-person pod for CPPIB (**\$550B+ AUM**), owning end-to-end ETL data mapping, relational schema design in Databricks and Airflow, and building PowerBI dashboards for senior portfolio managers, delivering timely insights that accelerated decision-making
- Re-architected ingestion with SQS/SNS into S3 and distributed PySpark across **1 TB+ daily data**, delivering **20% faster ingestion**, **40% lower latency**, and **15% higher reporting throughput**.
- Built VBA automation plugin integrating S&P Global and Visible Alpha data feeds into Excel financial models, performing fiscal/calendar year transformations and pushing cleaned outputs to the Valsys database for KPI dashboard refresh-reclaiming **25 hrs/week** and reducing manual errors by **70%**.
- Defined and enforced data SLAs and quality validation rules across **3+ client workstreams**, achieving zero reporting deadline misses and eliminating stakeholder escalations.

LTIMindtree | *Senior Data Engineer*

Jul 2020 - Oct 2023

- Led 3-person pod through full delivery lifecycle-sprint planning, code reviews, QBRs, governance audits-achieving **100% on-time delivery** across **5+** enterprise clients.
- Spearheaded migration of legacy SAP workflows to AWS Glue with Oracle-to-Snowflake transfers, cutting infrastructure and licensing costs by **~50%**.
- Optimized SAP BODS ETL pipelines integrating ECC, HANA, SQL Server, and Oracle into a centralized BI warehouse, compressing reporting from days to hours (**80% faster execution**).
- Automated **5+ recurring data-mapping workflows**, reclaiming **1.5 FTEs annually** and eliminating error-prone manual handoffs.
- Administered SAP BODS environments-job servers, patching, compliance upgrades-under enterprise governance frameworks, reducing system downtime by **20%** and maintaining continuous audit readiness.

EDUCATION

San Diego State University | *MS, Big Data Analytics* (GPA: 4.0 / 4.0 GPA)

Aug 2024 - May 2026

Indian Institute of Information Technology Jabalpur | *BTech, Computer Science*

Aug 2016 - May 2020

PROJECTS

AI-Powered Tax Advisory Platform | allabouttax.in | *AllAboutTax*

Dec 2025 - Present

- Architected a full-stack AI tax assistant using Next.js 16, React 19, and TypeScript, integrating OpenAI Assistants API with persistent thread-based conversation memory in PostgreSQL and Vercel AI SDK real-time streaming for complex tax guidance with cited legal references.
- Designed a freemium infrastructure with NextAuth.js OAuth, role-based rate limiting (RBAC) enforcing daily usage caps via Prisma ORM, and a Sanity headless CMS with Incremental Static Regeneration (ISR) delivering sub-second blog content updates without redeployment.

Composite Country-Level AI Scoring | aisi-bda600.vercel.app | *AI Supremacy Index*

Jan 2025 - May 2025

- Engineered a 7-dimension composite AI resilience scoring system, building reproducible ETL pipelines that integrated **31 heterogeneous CSV sources** into 8 analysis-ready datasets (**53,700+ rows**) with log-normalized scoring, coverage-weighted aggregation, and cross-dataset country name harmonization across **195 countries**.
- Built **25+ data visualizations** and engineered JSON extraction scripts powering **15+ interactive React/TypeScript** chart components (treemaps, bump charts, heatmaps, racing bar charts) and a Three.js 3D globe with **~70 collaboration arcs** and interactive country selection.

Predictive Accident Analysis | <https://trafficsensai.onrender.com/> | *TrafficSensAI*

Aug 2024 - Dec 2024

- Built end-to-end ML pipeline using DBSCAN spatial clustering and Random Forest classification on **500K+ incident records**, predicting accident severity at **87%+ accuracy** and surfacing high-risk urban hotspots.
- Developed Flask REST API serving real-time predictions to a React frontend with integrated ArcGIS visualizations, enabling city safety planners to explore severity trends and temporal patterns.

TECHNICAL SKILLS

- Data Engineering & Cloud:** Databricks, Airflow, SAP BODS, AWS Glue, PySpark, Delta Lake, Kafka, Snowflake, BigQuery, AWS (S3, Redshift, EC2, Athena), Azure AI Foundry, GCP, MongoDB, PostgreSQL, Data Warehousing, ETL/ELT
- AI & MLOps:** PyTorch, Hugging Face, LangChain, OpenAIAPI, RAG, LLM Fine-tuning, Prompt Engineering, W&B, FastAPI, Flask, ChromaDB, Docker
- Programming & Tools:** Python, PySpark, SQL, TypeScript, JavaScript, React, Next.js, Git, Jira, Confluence, Tableau, PowerBI, Power Automate